Biomedicine & Pharmacotherapy 147 (2022) 112621

Contents lists available at ScienceDirect

Biomedicine & Pharmacotherapy

journal homepage: www.elsevier.com/locate/biopha

The antioxidant potential of different edible and medicinal mushrooms

Ruth W. Mwangi [a], John M. Macharia [b,*,1], Isabel N. Wagara [a], Raposa L. Bence [b]

[a] Department of Biological Sciences, Egerton University, Nakuru, Kenya
[b] Doctoral School of Health Sciences, Faculty of Health Science, University of Pécs, Hungary

ARTICLE INFO

Keywords:
Mushrooms
Medicinal properties
Antioxidant potential
Free radicals

ABSTRACT

Mushroom consumption has grown extraordinarily owing to their high nutritional value, desirable taste, and aroma. Mushrooms continue generating lots of interest chiefly in their consumption as food, as a cure for different ailments, as well as important goods for commerce throughout the globe owing to their dietary, antioxidant, and therapeutic values. Higher Ascomycetes and Basidiomycetes mushrooms have different properties with anticancer and immunological potential. They as well provide vital health benefits and display a wide-ranging continuum of pharmacological effects. The antioxidant activity of different mushrooms was reviewed for different radicals including DPPH, ABTS, OH, Nitrite, metals, and lipid peroxidation. The present review presents pharmacological activities of different species of edible and medicinal mushrooms. This review provides tangible evidence that these mushrooms are an excellent source of natural constituents and antioxidants with potential application in pharmaceuticals and in treating and managing different diseases.

1. Introduction

Mushrooms belong to the class Basidiomycetes and some to Ascomycetes and are a widely distributed food resource around the globe (Plate 1). They are consumed because of their nutritional benefits as well as their medicinal values since time immemorial. In addition, they have a unique flavor and taste and because of their nutritional value as food, they have played a big role to improve human nutrition [1]. Mushrooms provide nutrients including easily digested proteins, carbohydrates, vitamins, fiber, minerals, and antioxidants. In many countries, mushrooms are considered as constituents of gourmet cookery, particularly because of their distinctive flavour and are esteemed by people as a cookery additive [2]. They are regarded as a rare dish with extraordinary functional and nutritional benefits, and they are similarly received as nutraceutical nourishments, they have significant importance because of their organoleptic value, medical properties in addition to their economic importance [3]. Nevertheless, it is not easy to distinguish between the medicinal from edible mushrooms since most of the commonly found edible ones have medicinal values while several species with medical values are likewise edible [4].

Mushrooms are nutritionally resourceful food in addition to being a source of valuable compounds with physiological activities. The fruiting bodies as well as mycelia possess constituents with extensive antimicrobial effects. Mushrooms are excellent sources of naturally occurring antibiotics; glucans (Plate 2) on their cell wall are renowned for immunomodulatory effects, besides their many secondary metabolites expressed and produced by the mycelia inhibit bacterial and viral pathogenic species [5]. Furthermore, the mycelial exudates from the mushroom show activity against protozoa, for instance, *Plasmodium falciparum,* the parasite that causes malaria and other microorganisms [6].

Mushrooms can form excellent substitute sources of compounds with antimicrobial properties, predominantly secondary metabolites, including steroids, anthraquinones, terpenes, quinolones, and derivatives of benzoic acid, as well as a few primary metabolite sources such as oxalic acid, proteins, and peptides [6]. *Lentinus edodes* is an edible mushroom and possesses antimicrobial activity against some gram-negative and gram-positive bacterial pathogenic species [7]. Furthermore, this species of mushroom has a great nutritional significance because it has high protein content, with a significant amount of fiber and indispensable amino acids, low in fats although these fats have exceptional vital contents of fatty acids. Additionally, this edible mushroom species offers nutritionally important vitamin content (B complex, C, D, and E) [8]. *Russula virescens* has an appealing taste and is a good source of calcium, phosphorus, proteins, among other compounds. The mushroom also possesses antimicrobial and antioxidant

* Correspondence to: Doctoral School of Health Sciences, Faculty of Health Sciences, University of Pécs, Vörösmarty Mihály str. 4, 7621 Pécs, Hungary.
E-mail address: johnmacharia@rocketmail.com (J.M. Macharia).
[1] ORCID: https://orcid.org/0000-0002-7535-9478

https://doi.org/10.1016/j.biopha.2022.112621
Received 1 October 2021; Received in revised form 4 January 2022; Accepted 5 January 2022
Available online 10 January 2022
0753-3322/© 2022 The Authors. Published by Elsevier Masson SAS. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

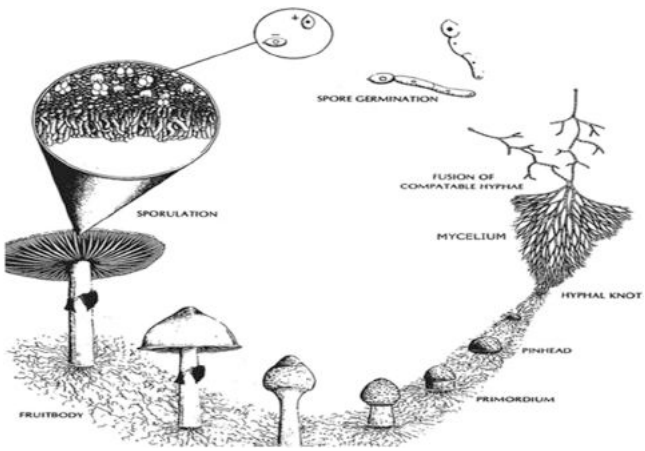

Plate 1. Stages of mushroom growth.

activities. *Boletus edulis* does not have fat or cholesterol and provides low calories in diet [9]. This mushroom also provides proteins and is a superb source of iron and dietary fiber, some minerals, for instance, zinc, manganese, and selenium, in addition to B-complex vitamins [9]. This mushroom is also an excellent good source of antioxidants and antimicrobials, therefore making it a choice food people would want to feed on [9]. Cordyceps contains many nutritionally important compounds such as different essential amino acids, vitamin K, and B-complex vitamins, as well as different nucleosides proteins, polysaccharides, and sterols [10]. This mushroom strengthens the immune system, reduces aging problems, promotes life longevity, and improves the functioning of the liver in hepatitis B patients [5]. In a most recent citation, it has been established in both a systematic review and a meta-analysis study that large quantities of mushroom consumption are highly correlated with lower risks of cancer development [11].

Therefore, mushrooms such as *Lentinula edodes, cordyceps, Russula virescens*, and *Boletus edulis* could be a superb source of numerous varied nutraceuticals and could be employed directly as human foods as well as in health promotion because the present bioactive constituents work synergistically [9]. Mushroom has proven as one of the most valuable candidates in the pursuit of bioactive substances [10–12]. This is because the fruiting bodies of mushrooms take a shorter production time; there may be rapid production of the mycelia using liquid culture while the culture medium could be adjusted for the production of maximum amounts of active substances [11–13].

The medicinal properties of mushrooms are attributable largely to their polysaccharide contents [17], exhibiting properties of the biological modulator, for instance, antibacterial, antitumor, and antiviral activities [18]. Considering the structural perspective, polysaccharides are an assorted cluster of organic macromolecules that occur in various entities [19]. They comprise repeated structural units, and they are monosaccharide residual polymers linked to one another using glycosidic bonds. Thus, they occur in form of nucleic acids and proteins and can carry more biological data because they can easily change their chemical structures depending on the prevailing conditions [20]. Studies associated with transcriptomics analysis and genomics of medicinal mushrooms such as *Lignosus rhinocerotis* have expanded greater understanding into the molecular biology of these mushrooms and created new ventures for more explorative studies at the genome level. Notably, the expressed genes in these medicinal mushrooms have significant bioactive potency important to human health [21].

Edible mushrooms provide outstanding sources of valuable bioactive metabolites and these compounds have a wide-ranging continuum of pharmacological uses inclusive of antioxidant effects [22]. Essentially, there exists a mounting interest to study biochemical and chemical characteristics of antioxidant ability of wild and cultivated edible mushrooms [23]. *Flammulina velutipes* is a dietetic supplement that provides exceptional benefits to hypertension patients, growing children as well as elderly persons. The biological activity of this mushroom species aids in reducing hypertension, cholesterol, and blood sugar, as well as having antithrombotic activities [24]. This mushroom species has exceptional benefits in human health [25]. The active polysaccharide, which comes in form of dietary fiber, is the edible part of *F. velutipes.* This fiber cannot be broken down using lytic enzyme nor is it digestible in the human digestive system. The fiber present in *F. velutipes* is water-soluble with the ability to combine with cholic acid or cholesterol through adsorption. Once adsorption occurs, the cholic acid amounts reverted to the liver are reduced while cholesterol metabolism and transformation to form cholic acid are increased. The concentration of cholesterol is decreased by disturbing lipid absorption within the ileum, thus, the blood lipid contents, as well as the likelihood of cardiovascular diseases, are drastically curtailed [25]. However, more studies need to be carried out on this mushroom species before it is recommended as a health supplement. This is because Mustonen et al., [26] reported potential deleterious outcomes on cardiac and skeletal muscles on mice when they were administered with 9 g/kg/d every day for five days. They as well reported the reduction of the lymphocyte and white blood cell counts on the same mice [26].

A free radical is a molecular species that can exist independently and has an unpaired electron making it highly reactive and unstable (Plate 3) [27]. Radicals can donate electrons to other molecules or accept electrons, thus demonstrating oxidation/reduction properties. The most prevalent free radicals associated with various diseases include hydroxyl radical, DPPH radical, superoxide anion radical, hydrogen peroxide, nitrite radical, and peroxy-nitrite radical. The highly reactive radicals occur in the cell membranes and nucleus and biologically damage molecules including lipids, carbohydrates, proteins, and nucleic acids. The free radicals also have the capacity of attacking vital macromolecules, which leads to damages to cells and tissues as well as disruption of

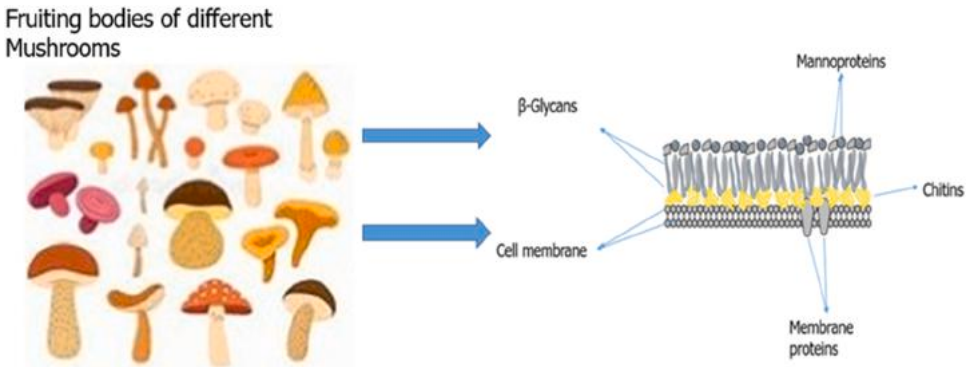

Plate 2. Glycan on mushroom cell walls and fruiting bodies.

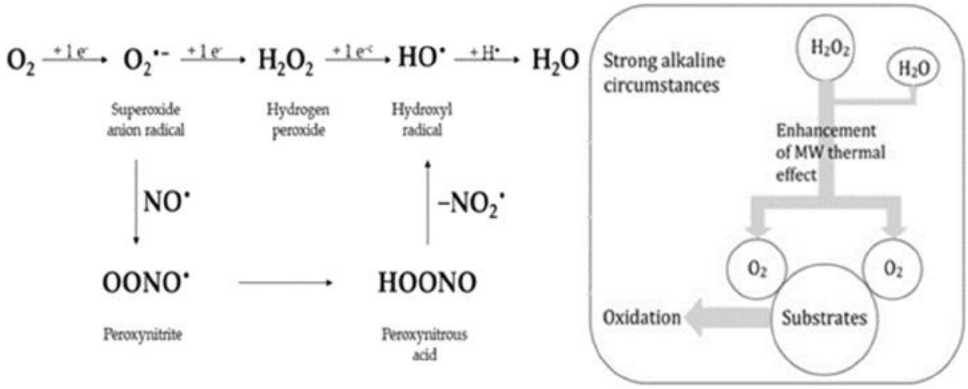

Plate 3. Generation of free radicals and how they are quenched.

homeostasis and metabolic processes [27].

Insufficient nutrition as a result of modern lifestyle as well as average longevity increase is the main reason why disease incidence throughout the globe is cumulatively growing [28]. Excessive contents of reactive oxygen species and imbalanced metabolism cause oxidative stress which generates a wide range of ailments including heart diseases, metabolic disorders, severe neural problems, for instance, Parkinson's and Alzheimer's disorders, some cancers, and premature aging [28]. Reactive oxygen species are not only produced internally, within the body of the organism, but also by several external sources such as ionizing radiation, ultraviolet light, inflammatory cytokines, environmental toxins, and chemotherapeutics [29].

In addition to the endogenous mechanisms of antioxidant defense of organisms, their nutritional intake is another significant antioxidant source that potentially contributes to oxidative homeostasis [30]. Foods containing antioxidants or antioxidant supplements (e.g. most edible mushrooms) might enable an organism to moderate oxidative injuries in addition to protecting food quality through averting oxidative deterioration. Antioxidants are used in the production and packaging of foods, as well as in healthcare, cosmetics, and anti-aging products, and here mushrooms find practical applications [31].

Wild mushrooms generally produce higher secondary metabolite concentrations when compared to popularly cultivated mushrooms [32]. The lower concentrations of secondary metabolites exhibited in cultivated edible and medicinal mushrooms might reflect the selection of mushroom cultivar and cultivation practices that greatly favor total volume and yield without necessarily considering the quality of secondary metabolites [33]. Precisely, the substrates used in mushroom cultivation may not provide all the necessary nutrients as well as climatic and environmental influences may contribute to these differences. Additionally, the use of irrigation and fertilizers may moderate the naturally occurring environmental stress which may trigger the mushrooms not to produce the secondary metabolites since they are mostly produced in response to stress.

Medicinal and edible mushrooms are found all over the world, on all continents. However, most of the species occur in subtropical humid forests as well as in temperate parts of North America, Europe, South America, Africa, Asia and Australia. In terms of production, China is the biggest producer and consumer of mushrooms [34]. The current work reviews the antioxidant activities of different edible and medicinal mushrooms. The review provides a critical synopsis and is intended for promoting further study and development of mushrooms for medicinal and health benefits.

2. Methods

A systematic search of relevant literature was carried out by searching the Web of Science, Ovid, BMC Springer, Elsevier, Embase, and MEDLINE databases for satisfactory screening and selectivity. Google Scholar was in addition used to supplement information from other articles relevant to our study. Importantly, articles from Google Scholar were verified for authenticity before being used in the present review. Lists of references were further scrutinized to identify additional related studies. Key terms 'edible/medicinal mushrooms' were combined with MeSH terms "antioxidant", "polysaccharides", "pharmacological activities", and "scavenging for radicals" to get relevant papers.

3. Phyto-constituents present in different edible mushrooms

Mushrooms are classified as vegetables in the United States Department of Agriculture food categories and have numerous nutritive properties like plant-derived foods. However, they are neither animal nor plant species but rather belong to the Fungi kingdom [11]. Phytochemicals are important bioactive elements meant for sustaining and promoting wellbeing and are present in various foods including mushrooms and medicinal plants [35]. Phytochemicals generally include anthocyanidins, phytoestrogens, carotenoids, terpenoids, phytosterols, limonoids, polyphenols, glucosinolates, isoflavonoids, and flavonoids. They have a remarkable impact on the medical field as they provide benefits such as preventing, managing, and treating ailments and different physiological conditions [36]. Free radicals largely originate from oxygen commonly known as reactive oxygen species (ROS) as well as from nitrogen generally called reactive nitrogen species (RNS). Radicals are generated in the human body by different endogenous systems, or when body tissues are exposed to various pathophysiological states or physicochemical conditions. These radicals can harmfully alter DNA, proteins, and lipids, and are associated with premature aging and various diseases affecting the human body [37].

3.1. *Active biomolecules present in edible and medicinal mushrooms*

A wide range of edible/medicinal mushrooms is identified to demonstrate biomolecules with biological activities. Generally, it is acceptable that mushroom extracts comprise enormous constituents, with each constituent producing specific effects in biological systems [38,39]. Active biomolecule substances present in mushroom fruiting bodies include flavonoids, phenolics, polysaccharides, glycosides, phenols, tocopherols, ascorbic acid, organic acids among other compounds (Table 1) [40,41]. In addition, mushrooms have been established to be important sources of ergocalciferol (vitamin D_2) upon exposure to ultra-violet radiation [11,42]. Vitamin D exists in two common forms, ergocalciferol and cholecalciferol (vitamin D_3) and it is a known pro-hormone largely associated with significant physiological functions such as bone mineralization, immune modulation, and calcium & phosphorus homeostasis among others. These two forms of vitamin D are involved in similar enzymatic hydroxylation activities to become active biomolecules [43].(Table 2).

Mushroom antioxidants exhibit their protective activities at diverse

Table 1
Compounds in edible/medicinal mushrooms responsible for antioxidant activity.

Mushroom Species	Compounds present	Type of activity	Reference
Agaricus arvensis	Phenols	DPPH radical scavenging, reducing power	[51]
Agaricus bisporus	Gallic acid, phenols, Flavonoids, Tocopherols, *p*- polysaccharides, hydrobenzoic acid derivatives, Pyrogallol	Reducing power, DPPH radical scavenging, lipid peroxidation inhibition, scavenging of superoxide radicals	[52]
Agaricus brasiliensis	Polysaccharides, phenols	Reducing power, lipid peroxidation inhibition, scavenging of OH & DPPH radicals	[53]
Agaricus comtulus	Phenols	β-carotene bleaching inhibition	[54]
Agaricus campestris	Phenols	Reducing power	[55]
Agaricus lutosus	Phenols	β-carotene bleaching inhibition	[54]
Agaricus macrosporus	Agaricoglycerides	Anti-inflammatory	[54]
Agaricus romagnesii	Phenols	Lipid peroxidation inhibition, reducing power	[56]
	Polysaccharides	Reducing power	[57]

(*continued on next page*)

Table 1 (*continued*)

Mushroom Species	Compounds present	Type of activity	Reference
Armillaria mellea			
Auricularia auricular	Phenols, glucan	Reducing power	[58]
Antrodia camphorata	Polysaccharides, phenols, ascorbic acid, γ-tocopherol, diterpenes	Chelating for Ferrous ion, scavenging for DPPH radical, lipid peroxidation inhibition	[59]
Boletus edulis	Organic acids, phenols e.g., flavonoids, polysaccharides	Reducing power, lipid peroxidation inhibition, chelating for Ferrous ion, scavenging for DPPH radical	[60]
Cantharellus cibarius	Polysaccharides, flavonoids, pyrogallol	Scavenging for DPPH radical	[61]
Calvatia gigantean	Calvacin, phenolic compounds, polyunsaturated fatty acids	Scavenging for DPPH radical, cupric reducing antioxidant activity	[62]
Cordyceps Sinensis	Polysaccharides, exopolysaccharides, protein complexes	Ferric reducing potential	[63]
Cyathus africanus	Diterpenoids (11-O-acetylcyathatriol, neosarcodonin, and cyathatriol	inhibition against production of nitric oxide	[64]
	Phenols, Polysaccharides, Flammulin	Lipid peroxidation inhibition, scavenging of DPPH and OH, β-carotene bleaching inhibition	[65]

(*continued on next page*)

Table 1 (*continued*)

Mushroom Species	Compounds present	Type of activity	Reference
Flammulina velutipes			
Ganoderma applanatum	Phenols, β-glucans	Scavenging for DPPH radicals	[66]
Ganoderma lucidum	Phenols, carboxymethalated polysaccharides, pepetides and polysaccharides, ganoderiol, ganoderic acids, ganodermanontriol	Scavenging of superoxide, DPPH and OH radicals	[67]
Hericium erinaceous	Phenols, steroids, erinacines, hericenons hericerins, resorcinols, mono-terpenes, diterpenes	Lipid peroxidation inhibition, scavenging for DPPH radical	[68]
Lentinus edodes	Chitosan, polysaccharides, phenols, *p*-hydroxybenzoic acid, mannoglucan, lentinan, glucan,	Scavenging for DPPH & OH radicals, chelating for Ferrous ion, lipid peroxidation inhibition	[3,69]
Lignosus rhinoceros	Phenolic compounds, Polysaccharide-protein	scavenging against superoxide anion radical	[70]
Morchella esculenta	Extracellular polysaccharide, phenols, β-1,3-D-glucan, galactomannan, heteroglycan	Scavenging for ABTS, DPPH & OH radicals, lipid peroxidation inhibition, chelating for Ferrous ion	[71]
	Heteropolysaccharide acid, laccase, polysaccharide-protein complex	Scavenging against ABTS, DPPH superoxide, and hydroxyl radicals	[72]

(*continued on next page*)

Table 1 (*continued*)

Mushroom Species	Compounds present	Type of activity	Reference
Pleurotus eryngii			
Pleurotus florida	Phenols, intracellular polysaccharides	Scavenging against superoxide radical	[73]
Pleurotus ostreatus	α-tocopherol, β-carotene, cinnamic acid, flavonoids, phenols	Chelating for Ferrous ion, lipid peroxidation inhibition, scavenging for superoxide, OH, and DPPH radicals	[74]
Pleurotus sajor-caju	Phenols	Lipid peroxidation inhibition, scavenging for DPPH radical	[75]
Polyporus squamosus	Tocopherols, phenols	Scavenging for DPPH radical	[76]
Polyporus tenuiculus	Phenols	Chelating for Ferrous ion, scavenging for DPPH radical	[77]
Russula delica	Phenols, catechin, tocopherols	Inhibition for β-carotene bleaching, chelating for Ferrous ion, scavenging for DPPH & superoxide radicals	[78]
Russula virescens	Tocopherols, phenols	Lipid peroxidation inhibition	[79]
	Tocopherols, Phenols	Scavenging for superoxide radical	[80]

(*continued on next page*)

Table 1 (*continued*)

Mushroom Species	Compounds present	Type of activity	Reference
Verpa conica			
Volvariella volvacea	Phenols	Scavenging for DPPH radical, lipid peroxidation inhibition	[81]
Xylaria hypoxylon	Lectin glycoproteins	chelating for Ferrous ion	[82]

Table 2
Mushrooms with antimicrobial properties.

Antifungal	Antibacterial	Antiprotozoal	Antiviral
Sparassis crispa	*Flammulina velutipes*	*Clitocybe illuclenia*	*Coprinus micaceus*
Oudemansiella radicata	*Polyporus squarnosus*	*Plasmodium gallinaceum*	*Panaeolus sphinctrinus*
Lentinus lepideus	*Psathyrella* sp		*Agaricus campestris*
Lentinu edulis	*Tricholoma* sp.		*Lentinus edodes*
Coprinus comatus	*Trametes* sp.		*Armillaria mellae*
Boletus edulis	*Agaricus bisporus*		*Russula virescens*

stages in the process of oxidation as well as using diverse mechanisms [44,45]. There exist two key kinds of antioxidants found in mushrooms, viz., primary (free radical scavengers thus chain-breaking antioxidants) in addition to secondary or defensive [46–48]. Secondary antioxidants are produced when metals are deactivated, lipid hydroperoxides are inhibited or broken down when primary antioxidants e.g. singlet oxygen ($^1O_2$) are regenerated or quenched among other processes. Other mushroom compounds exhibiting antioxidant properties function as cell signals and/or inducers, which leads to gene expression alterations, resulting in enzyme activation thus eliminating reactive oxygen species [45,49]. Dietary mushrooms such as *P. ostreatum* have been demonstrated to evoke a hypocholesterolemic sequel and prevent lipid peroxidation, a mechanism strongly associated with oxidative stress [50].

4. The scavenging potential of mushrooms

4.1. Antioxidant activity

The oxidation process is very essential to various living organisms for energy production necessary for the physiological processes. The scavenging of freely available radicals is a technique for lipid oxidation inhibition generally used for evaluating antioxidant potential [83]. Free radicals derived from oxygen are injurious to the health of humans since they are extremely reactive particles as they contain unpaired electrons [84]. Throughout oxidative stress, freely available radicals attack cells, tissues, and organs of organisms, along with their proteins RNA, DNA. The ultimate consequence is cell injury, apoptosis, and necrosis leading to body disorders and illnesses such as cardiovascular diseases, diabetes, abnormal aging, and various forms of cancers [85]. The inhibitive potency of mushrooms fungal species against premature mortality is closely linked to their large quantities of potent antioxidants such as ergothioneine and glutathione [86].

4.2. 1,1-Diphenyl-2-picrylhydrazyl (DPPH) radical scavenging

The assay on DPPH radical is useful in predicting antioxidant activities by inhibiting lipid oxidation. Therefore, DPPH radical scavenging determines the extent to which free radicals are scavenged. The DPPH radical scavenging method is popularly used because of the comparatively short time needed to carry out the analysis [87].

Results from various studies vary and this may be due to different environmental conditions, cultivation methods, stages of development when harvested, and genetic variation among the strains. Abul Hasnat et al. [88] reported that the ethanol and aqueous extracts of *Russula virescens* demonstrated scavenging potential against DPPH radicals in a dose-dependent fashion. The aqueous extract showed a scavenging potential of 12.67–81.13% while that of ethanol extract showed an activity of 5.30–52.62%. The concentrations of 0.125–2.0 mg/mL of *R. virescens* extract were tested and the effective concentration (EC_{50}) levels found were 0.22 mg/mL for the aqueous extract while the ethanol extract EC_{50} was found to be 1.90 mg/mL. When these EC_{50} values were compared with ascorbic acid which served as a positive control (a concentration of 2.0 mg/mL showed 95.90% activity), the scavenging potential of the two extracts was significantly high. Yap et al. [89] reported a DPPH scavenging ability of *Lignosus rhinoceros* and the concentrations ranged between 1 and 16 mg/mL however, the DPPH scavenging potential was dose-dependent [90].

Mau et al. [91] reported scavenging effects ranging between 96.3% and 99.1% using *Antrodia camphorata* methanolic extracts at a concentration of 2.5 mg/mL. *Dictyophora indusiata* methanolic extract at a concentration of 6.4 mg/mL showed 92.1% DPPH radical scavenging activity, while 63.3–67.8% scavenging activity was recorded with extracts of other mushroom specialties prepared in methanol [92]. At a concentration of 6.4 mg/mL, the *Pleurotus ostreatus* methanolic extract showed 81.8% scavenging activity against DPPH radicals, while 42.9–69.9% scavenging activities were obtained using extracts of

various commercial mushrooms [93]. Furthermore, at a concentration of 1 mg/mL, extracts of *Auricularia auricula-judae* (commonly known as black and red ear mushrooms) prepared in methanol showed 100% DPPH radical scavenging activity, whereas the *Tremella fuciformis* (snow mushrooms) methanol extract showed 94.5% DPPH radical scavenging activity at a concentration of 0.4 mg/mL [13].

In another study, Menaga et al. [94] conducted tests using methanolic extract of various edible mushrooms at a concentration of 6.4 mg/mL. *Pleurotus geesteranus* showed 81.8% scavenging potential against DPPH radicals, while *Lentinus edodes, Flammulina velutipes* in addition to oyster cap mushrooms recorded 43∼70% scavenging activity against similar radicals [94]. Mushrooms with medicinal properties could provide substances resistant to oxidation and they could as well act as scavengers of the freely available radicals. Phenolic substances represent the major cluster of phytochemical constituents as well as phenolic compounds with the ability to inhibit low-density lipoprotein (LDL) oxidation [95]. The results show that medicinal and edible mushrooms can be good antioxidants [96]. In an entirely different research, Hussein et al. [97] demonstrated that the methanol extract of *Auricularia auricular-judae* possessed significantly strong scavenging potential of 93.33% at an EC_{50} of 0.08 mg/mL, *Polyporus tenuiculus* showed 90.71% activity whereas *Polyporus conchatus* showed 46.53% activity and was the weakest with an EC_{50} of 0.3 mg/mL. Other researchers have reported that edible mushrooms show antioxidant activity against the DPPH radicals [68,69]. The results obtained show that mushrooms with medicinal properties possess free radical scavengers and inhibitors, which act possibly as primary antioxidants. They possibly interact with the freely available radicals, principally the radicals of peroxide, which act as the autoxidation fat chain propagators, thus terminating the chain reaction [27,71].

In a different study, Elmastas et al. [80] reported that extracts of methanol from different species of mushrooms displayed strong scavenging potential against DPPH free radicals. *Lepista nuda* showed 91.3 mg/mL, *Russula delica* 86.1 mg/mL, *Polyporus Squamosus* 82.8 mg/mL, *Pleurotus ostreatus* 81.3 mg/mL *Agaricus bisporus* 77.5 mg/mL, *Verpa conica* 75.7 mg/mL, and *Boletus badius* 68.7 mg/mL. The results obtained in this study showed that extracts of methanol from the different species of mushrooms produce an appreciable activity on scavenging the freely available radicals [80]. These results showed that mushroom extracts of methanol could act as scavengers of free radicals, where they could act as the principal antioxidants. Wild edible mushroom methanolic extracts might interact with free radicals that are the primary autoxidation fat chain initiators, thus terminating the chain reaction [45].

Hung, [100] worked on five different species of mushrooms and reported their antioxidant potential. They reported the antioxidant activity of the free phenolic extracts where *V. Volvacea* showed the highest activity of 82.9%, followed by *L. edodes* which recorded 75.8%, then *P. ostreatus* which recorded 45.4%. *G. lucidum* recorded 40.7% while the lowest activity was exhibited by *A. polytricha,* which showed a 21.1% activity [100]. The bound phenolic extract of *G. lucidum* exhibited the highest antioxidant potential of 7.5% while *L. edodes* recorded 6.7% capacity, *V. volvacea* showed 4.2% potential, *P. ostreatus* recorded 1.8% whereas *A. polytricha* showed the least potential of 0.7%. They further reported that the antioxidant potential of the free phenolic extracts was higher in comparison to that of the bound phenolic extracts. The excellent antioxidant potential of mushroom phenolic extracts proves that mushrooms are not only consumed as nutritious foods but also used as functional foods and pharmaceutical medicines [101,102].

Keleş et al. [103] reported the scavenging potential of some mushroom species. They reported the methanolic extract of *Boletus edulis* and *Suillus luteus* as showing the greatest scavenging potential (93.18, 97.96%, respectively, at a concentration of 25 mg/mL). However, *Hydnum repandum* recorded the lowest potential among all the mushroom species they worked on and recorded 10.17% activity [103]. The scavenging activities of methanol extracts from edible species of mushrooms as well as the standards against the DPPH free radicals were as follows: *Lepista nuda* showed the highest activity of 91.3%, *Russula delica* showed significantly high activity of 86.1%, *Polyporus squamosus* was similarly high 82.8%. *Pleurotus ostreatus* showed appreciably high activity of 81.3%, *Agaricus bisporus* showed 77.5% activity, *Verpa conica* recorded 75.7% activity, while *Boletus badius* showed scavenging activity of 68.7% [103]. All the concentrations of the extracts were prepared at a level of 180 mg/mL [104]. Keleş et al. [103] further demonstrated that scavenging activity against DPPH radicals reduced following this order: *P ostreatus (*86.35*), L. nuda (*85.61%*), A. bisporus (*67.86%*), P. squamosus (*43.30%*), R. delica* (37.10%) but showed some slight differences from the study above. All the extracts were prepared at a concentration of 25 mg/mL. The antioxidant potential of plant tissues is related to the effect of enzymes that scavenge against free radicals. These enzymes include (catalase, peroxidase superoxide dismutase, among others, as well as the antioxidant compounds, mainly ascorbic acid, tocopherol, phenolic substances, and carotenoids. There is a growing interest in evaluating the antioxidant effects of direct mushroom extracts [105].

Manimaran et al. [106] worked with Zinc oxide nanoparticles of *Pleurotus djamor* against DPPH radical scavenging activity and reported LC_{50} value of 428.35 μg/mL showed 59% inhibition of the radical. This activity was significantly high compared to the standard vitamin C which recorded 69% inhibition at an LC_{50} value of 368.43 μg/mL.

4.3. $ABTS^+$ radical scavenging

The assay on ABTS assesses the comparative capacity of antioxidants to scavenge the ABTS produced in the water phase when compared with a Trolox standard (water-soluble vitamin E). The ABTS is produced when a strong oxidant agent, for instance, potassium persulfate or potassium permanganate reacts with the ABTS-based salt [107].

Results from various authors show that both ethanol and aqueous extracts possess significant scavenging potential against $ABTS^+$ radicals. Hasnat et al. [88] reported the scavenging activity of *Russula virescens* to range between 3.28% and 87.12% and 14.43–96.45% for ethanol and aqueous extracts, respectively. The concentration of the extract ranged between 0.125 and 2.00 mg/mL. The EC_{50} values were 0.63 and 0.44 mg/mL for ethanol and aqueous extracts, respectively. The *R. virescens* ethanolic extract was considerably less effective (maximum 87.12%) when compared to that of the standard agent ascorbic acid which showed 99.48% scavenging activity both at a concentration of 2.0 mg/mL [88].

Zhenyu et al. [108] used different samples to study the antioxidant ability of mushroom extracts. They observed that the scavenging activity against $ABTS^+$ radical rose consistently as the sample concentration was increased. At a 20 mg/mL concentration level, the highest rate of radical scavenging of the extracts of mushroom species ranged between 18.54% and 100%. Amongst all the extracts, the *Russula vinosa* Lindblad acid extracts displayed the maximum antioxidant activities. The majority of the acidic extracts showed greater scavenging activities compared to the alkaline and aqueous extracts [109].

The ethanol extract of *Omphalotus olearius* at a level of 1 mg/mL exhibited significantly high scavenging activity against the $ABTS^+$ radicals of 88.01% [23], while the ethanol extracts of *Rhizopogon roseolus* and *Morchella esculenta* var. *rigida* demonstrated 83.13% and 87.07% respectively. At a concentration of 40 μg/mL, extracts of *Morchella conica* fruiting bodies extracted using methanol showed 78.66% scavenging activity against $ABTS^+$ radicals [19]. At a concentration of 0.14 mg/mL, extracts of fruiting bodies of *Amanita cesarea* and *Boletus edulis* prepared in methanol showed 92.0% and 85.8% respectively, scavenging activity against $ABTS^+$ radicals [47].

Gargano et al. [110] worked with methanol extracts made from various mushroom species and observed that the antioxidant potential of aqueous extracts was dependent upon the concentration used; however, lower doses were required to eliminate all the $ABTS^+$ radicals.

Aqueous extracts prepared using *Boletus edulis, Lentinus edodes,* and *Amanita cesarea* exhibited a very high $ABTS^{+}$ scavenging capacity that ranged between 85.8% and 92% at significantly low concentrations of 0.14 mg/mL. The aqueous extracts of *Morchella esculenta, Agaricus bisporus, Lactarius deliciosus* and *Pleurotus* sp. ware reported having an intermediate radical scavenging capacity ranging from 26% to 60% while the least potential was observed with the aqueous extract of *Cantharellus cibarius,* which was 10.6% [110]. Further, Manimaran et al. [106] worked with Zinc oxide nanoparticles of *Pleurotus djamor* and reported that they possessed an active free radical quenching ability, and the inhibition percentage was concentration dependent. The Zinc oxide nanoparticles of *P. djamor* resulted in the LC_{50} value of 404.70 μg/mL with a 59.30% inhibition, while the standard vitamin C exhibited 73.20% of inhibition with the LC_{50} value of 332.82 μg/mL [106].

4.4. Scavenging activity against hydroxyl (OH)

Hydroxyl free radicals (OH) form a critical oxidizing species with a great ability to attack and oxidize molecules within their locality in an effort of balancing the configuration of the unpaired electrons. Therefore, owing to their high oxidizing power, hydroxyl radicals react aggressively with oxidative systems [111].

Amongst the ROS, the most reactive radical is hydroxyl because it can trigger severe injury to adjacent biomolecules. Yang et al. [112] reported that the hydroxyl radical can trigger severe injury to neighboring biomolecules within the body that results in cell damages which cause premature aging, the development of cancerous cells, and various other illnesses. Elimination of hydroxyl radicals is perhaps one of the most effective means to defend against oxidative impairment of cells in the body. Thus, the scavenging effects against the hydroxyl radicals are perceived as the most vital mechanisms of autoxidation. Superoxide and hydrogen peroxide molecules can cause oxidative damage to neighboring biomolecules not directly through generating the OH radicals [112].

Hussein et al. [97] showed the hydroxyl scavenging effects from unpurified polysaccharides obtained from edible mushrooms and they displayed a dose-dependent pattern. Amongst all the extracts of mushrooms, the aqueous extracts of *Dictyophora indusiata, Hohenbuehelia serotina,* and *Hypsizygus marmoreus* showed the highest scavenging activities against hydroxyl radicals, while the scavenging potential reached 100%. At 2.0 mg/mL concentration, the aqueous extracts scavenging potential against the OH were all above 50% except for the extract of *Auricularia auricula*, which recorded 39.5% [98]. It was observed that alkaline extracts exhibited lower scavenging activities when compared to aqueous extracts. The *Auricularia auricula* and *Russula vinosa Lindblad* alkaline extracts displayed the least (EC_{50} 27.85 ± 0.56 mg) and the best (EC_{50} 50.60 ± 0.55 mg) scavenging ability against the hydroxyl radicals [108]. Two different mechanisms of autoxidation against the OH radicals are involved: the first is direct hydroxyl radical scavenging while the second mechanism is directly suppressing the generation of hydroxyl radicals [113].

Hasnat et al. [88] reported that the extracts of *Russula virescens,* an edible mushroom, displayed scavenging effects against hydroxyl radical in a dose-dependent manner. At 0.125–2 mg/mL concentrations, the scavenging potential ranged between 39.69% and 63.71% for *R. virescens* ethanol extract and 38.33–71.21% for *R. virescens* aqueous extract. Ascorbic acid, which served as the standard, exhibited a scavenging potential of 86.97% at a concentration of 2 mg/mL. It was observed that the activity of ascorbic acid was significantly superior when compared to the activities of ethanol and aqueous extracts of *R. virescens.* Superoxide anions form antecedents to free radicals that are active and display some ability to react with biological macromolecules and thus trigger soft tissue injury [114]. It has been reported that superoxides sometimes directly initiate the peroxidation of lipids. The superoxide anion performs a vital function during the creation of other reactive oxygen species including singlet oxygen, hydroxyl radical, and hydrogen peroxide, which can induce oxidative impairment on DNA, proteins, and lipids [102]. In another research, *R. virescens* ethanoic extract displayed scavenging activity against superoxide radicals, ranging between 3.49% and 53.93% at 0.125–2mgmL^{-1} concentrations [88]. Ascorbic acid as the positive control exhibited a scavenging potential of 76.13% at a concentration of 2mgmL^{-1}. This was considerably superior when compared to the activity of the *R. virescens* ethanolic extract [102].

Zou et al. [115] reported that the scavenging ability of antioxidant polysaccharides extracted from the fruiting bodies of *Auricularia auricula* showed a dose-dependent activity. For instance, at a dose ranging between 0.2 and 1.0 mgmL^{-1}, crude polysaccharides extracted from the fruit bodies of *Auricularia auricula* showed significantly stronger scavenging activities compared to the standard butylated hydroxytoluene, considering the EC_{50} of 0.38mh/mL in the mushroom while that of the standard was (0.56 mg/mL) [115]. In comparison with the fruit bodies of *Gloeostereum incarnatum,* the polysaccharides obtained from the fruit bodies of *Auricularia auricula* showed higher scavenging potential against the OH radicals at the same concentrations [17]. Their results suggest that *Auricularia auricula* polysaccharides offer better natural antioxidants than butylated hydroxytoluene in scavenging hydroxyl radicals [47]. Further, well documented biochemical research has demonstrated that the extracts from the fruiting bodies of two edible mushrooms, *Coprinus comatus* and *C. truncorum* are highly effective in neutralizing OH radicals. Of the two mushroom species, *C. truncorum* exhibited the highest activity of anti-OH radical potency [116]. Thus, these results demonstrate that extracts of mushrooms, through electron-donating or/and hydrogen action, could avert radical species which are reactive from getting to biomolecules, for instance, proteins, amino acids, DNA, lipoproteins sugars in addition to polyunsaturated fatty acids on the vulnerable food and biological systems [117].

4.5. Metal chelating ability

Metals for transition are believed to catalyze the preliminary radical formation. Chelating agents can steady transition metals in living organisms as well as inhibit the generation of free radicals, therefore decreasing oxidative damage induced by free radicals [118]. To evaluate the autoxidation effect of any extract in a better way, its chelating effect is always assessed against ferric (Fe^{2+}) ions. Zou et al. [115] reported that *Auricularia auricula* polysaccharides possessed higher Fe^{2+}-chelating activity compared to the standard butylated hydroxytoluene in a dose-dependent fashion. The EC_{50} value (0.43 mgmL^{-1}) of *Auricularia auricula* polysaccharides having a ferrous chelating effect was significantly poorer than butylated hydroxytoluene, at 0.79 mg/mL [118]. They also reported that at similar concentrations, the chelating effect of *Auricularia auricula* polysaccharides on ferric ions was higher than the chelating activity of polysaccharides obtained from *Agaricus brasiliensis*, *A. bisporus, Ganoderma lucidum,* and *Phellinus linteus* fruiting bodies. Mau et al. [91] reported the methanolic extracts from *Antrodia camphorata* to chelate Fe^{2+} by 64.4–74.5% at a concentration of 5 mgmL^{-1}, while the chelating ability of *Agaricus blazei* mushrooms was excellent and was 98.6% at a concentration of 2.5 mgmL^{-1} [118].

Hasnat et al. [88] reported that the ethanol and aqueous extracts of *Russula virescens* displayed chelating potential in a concentration-dependent pattern. The scavenging potential ranged between 3.2% and 56.69% when using extracts prepared in ethanol and 5.7–68.10% when using extracts made in water all at 0.125–8 mgmL^{-1} concentrations. The EC_{50} values obtained were 6.82 and 3.63 mg/mL for extracts of ethanol and water respectively. Nevertheless, the chelating effect of EDTA, which served as the standard was higher (90.87% at a concentration of 0.1 mg/mL) compared to the two extracts. In a different study, when using a 1.6 mg/mL sample concentration of *Pleurotus geesteranus*, *Flammulina velutipes*, *Lentinus edodes,* and oyster cap mushroom, the chelating potential against ferric ions was found to range between 45.6% and 81.6% [47].

Menaga et al. [94] evaluated the chelating activity of the methanol extract from *Pleurotus florida.* They reported that the extract was able to chelate at 64% using a concentration of 500 μg/mL. They found the EC_{50} value of the same mushroom species to be 355 μg/mL concentration even though the chelating potential was concentration-dependent. Nonetheless, EDTA, a synthetic chelator for metals displayed an outstanding capacity of 81% at a similar dose. Likewise, Hussein et al. [97] indicated that extracts from wild edible mushrooms prepared in methanol showed higher chelating abilities against ferric ions when compared to the chelating abilities of cultivated mushrooms. Data from various authors indicate that wild and cultivated mushrooms possess bioactive molecules in addition to potential nutrients that have the capacity of preventing various diseases as well as improving human health [119].

The capacity to chelate metals is vital because it can decrease the transition metal concentrations, which could catalyze the generation of the initial small radical numbers that start the radical facilitated chain reactions autoxidation within the food and biological systems [120]. Pal et al. [121] reported that agents for ion chelating as well could prevent the hydroperoxide decomposition as well as Fenton reaction. Various studies have reported that mushroom extracts chelating capacity increases with an increase in concentration; it is dose-dependent [121].

4.6. Scavenging ability against nitrite

Exposure to nitrite/nitrate radicals is associated with adverse health effects including nausea, increased heart rate, abdominal pains, and headaches. Nitrite interacts with amines in medicines, proteinaceous food substances, as well as residual synthetic chemicals forming nitrosamines and is found in huge amounts in meats, roots in addition to leafy green vegetables. The nitrosamines formed are transformed into intracellular components, proteins, and diazoalkane that can increase the risk of developing cancerous cells. Martínez et al. [122] attributed the high scavenging potential against nitrite to the large quantities of phenolic substances naturally occurring in plants. Wang et al. [123] reported that the extract of *Pleurotus florida* prepared using methanol effectively inhibited sodium nitroprusside from generating nitric oxide. The EC_{50} value of the methanolic extract of *P. florida* displayed scavenging effects against nitric oxide at a dose of 510 μg/mL whereas vitamin C which served as a standard was effective at a dose of 240 μg/mL [123].

Sánchez [102] reported that the ethanol and aqueous extracts of *Russula virescens* exhibited a scavenging effect against nitrite in a concentration-dependent manner. The scavenging potential ranged between 28.601% and 41.94% for ethanol extracts and 35.21–53.98% for aqueous extracts at a dose ranging between 0.125 and 2mgmL^{-1}. The standard agent; ascorbic acid exhibited an EC_{50} value at a concentration of 0.45 mgmL^{-1}, while its scavenging potential was considerably superior (88.21% at a concentration of 2.0 mg/mL) when compared to the scavenging activities of the *R. virescens* extracts [124]. Wagay and Vyas [125] investigated the methanolic extract of *Morchella esculenta* mushroom for its scavenging activities against the production of nitric oxide (NO). The sodium nitroprusside could produce nitrite radicals at an optimum biological pH and mushroom extracts were found to have the capacity of inhibiting these radicals. A 100 μg/mL dose of the mushroom extract exhibited 49.04% scavenging activity, however, it was dose-dependent. A 100 μ g/mL of the ascorbic acid exhibited 63.02% inhibition, which was significantly higher when compared to the activity of the mushroom extracts [126].

Nitha et al. [127] investigated the *Morchella esculenta* ethanol-aqueous extract. They reported that the extract inhibited sodium nitroprusside from generating nitric oxide. They incubated various doses of the *M. esculenta* extracts with solutions of sodium nitroprusside in phosphate buffered saline and they found that there was a decrease of nitric oxide production, which showed a concentration-dependent pattern. Their results revealed that the mushroom extracts had significant scavenging effects against nitric oxide [127]. This investigation with the scavenging potential of extracts of *M. esculenta* against the radicals of nitric oxide indicated that this mushroom is an excellent scavenger against the radicals of nitric oxide. The nitric oxide radicals produced by sodium nitroprusside interacted with oxygen through a chemical reaction forming a nitrite [127]. The mushroom extract inhibited the formation of nitrite as it had a high affinity for oxygen. Therefore, it competed with oxygen for direct reaction with nitric oxide as well as inhibiting the synthesis of nitric oxide [124]. Notably, the production of NO is highly correlated with immunopathogenesis of inflammatory diseases such as vascular diseases and atherosclerosis. In this regard, it has been reported that methanolic extracts from the established edible mushrooms prevents production of NO via down-regulation of COX-2 and the Inducible nitric oxide synthase (iNOS) expression [128].

4.7. Lipid peroxidation

Lipid peroxidation can interrupt the membrane assemblage, which causes changes in permeability and fluidity, inhibits metabolic processes, and alters ion transport. When mitochondria are injured through lipid peroxidation, more reactive oxygen species are generated. Acharya et al. [129] found that methanolic extracts of *Antrodia camphorata* a medicinal mushroom exhibited outstanding antioxidant effects as shown by lipid peroxidation ranging from 5.32% to 5.78% at a concentration of 1.0 mgmL^{-1}. *Agaricus blazei* methanolic extract; another mushroom with medicinal properties displayed lipid peroxidation of up to 26%, at a concentration of 1.0 mg/mL, a significantly high antioxidant potential [129]. Nowacka, et al. [130] worked with four edible mushrooms and investigated their antioxidant potential against lipid peroxidation. This mushroom specialty included *Dictyophora indusiata, Grifola frondosa, Hericium erinaceus,* and *Tricholoma giganteum.* They used a concentration of 1.2 mg/mL and prepared the extracts using methanol as a solvent [130]. The extract of *D. indusiata* displayed an exceptional antioxidant potential of 72.26% against lipid peroxidation [131]. The extract of *G. frondosa* exhibited a moderately high antioxidant effect of up to 29.81% against lipid peroxidation, while *H. erinaceus* showed 48.45% of lipid peroxidation whereas *T. giganteum* displayed relatively high antioxidant effects of up to 67.02% against lipid peroxidation [92]. Extracts were made from four commercial mushrooms including *Pleurotus ostreatus P. cystidiosus, Lentinula edodes,* and *Flammulina velutipes* using methanol as a solvent. A concentration of 1.2 mg/mL of the methanolic extracts exhibited antioxidant effects ranging from 24.80% to 62.23% against lipid peroxidation as described by Mleczek et al. [92]. Gąsecka et al. [13] reported that ear mushroom extracts prepared in methanol as a solvent including silver, snow, jin, red, and black ear mushrooms exhibited relatively high antioxidant effects ranging from 57.7% to 71.5% against lipid peroxidation. These extracts were prepared to a concentration of 1.0 mg/mL [129]. Additionally, Karaman et al., [132] worked with methanol extracts of *G. lucidum G. applanatum*, and *L. sulphureus* as well as $CHCl_3$ extracts of *M. giganteus* and *G. lucidum* and reported very high inhibition against lipid peroxidation. The methanol extracts of these mushrooms reduced Fe^{2+} /ascorbate-induced lipid peroxidation in liposomes as an oxidizable substrate, and the results were dose dependent.

In most of the mushroom species that demonstrated scavenging potential, there was a linear association between the total content of phenolic compounds found in the extracts of the mushrooms and their antioxidant potentials [133]. This ratifies that medicinal and edible mushrooms, both wild and cultivated possess potent as naturally occurring antioxidants because of the capacity of their phenolic compounds to prevent lipid peroxidation [134].

5. Conclusion and future recommendation

Mushrooms have antioxidant bioactive compounds such as flavonoids, phenolics, polysaccharides, glycosides, phenols, tocopherols,

ascorbic acid, organic acids among other compounds that exhibit inhibitory and immunological potency. These biomolecules play vital roles such as scavenging free radicals in the body and thus inhibiting carcinogenesis. This review further demonstrates that mushrooms are largely used as food and supplements because of their abundant levels of dietary fiber, proteins, minerals, and vitamins. Of significance, mushrooms synthesize numerous chemically assorted substances with a comprehensive range of physiological functions. Phenolic biomolecules are therefore naturally occurring substrates for oxidative enzymes, which are abundant in large amounts in mushrooms. Mushrooms have significant health benefits as demonstrated in this review that should be explored to improve the quality of life. Finally, more studies should be carried out to gain a deeper insight into the antioxidant activities of different mushroom species and how to formulate the active metabolites into food supplements and drugs.

CRediT authorship contribution statement

Ruth W. Mwangi performed Conceptualization; Methodology; Writing original draft; Formal analysis; Final approval of the version to be submitted. **John M. Macharia** performed Data curation; Investigation; Visualization; Final approval of the version to be submitted. **Isabel N. Wagara** carried out Investigation; Reviewing the final draft and editing; Final approval of the version to be submitted. **Bence L. Raposa** was involved in Supervision; Project administration; validation; Data curation; Final approval of the version to be submitted.

Author contribution

All the authors contributed significantly to the development, review, and editing of the manuscript. All authors approved the submission of the manuscript.

Funding

This work did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Conflict of interest statement

The authors declare no conflict of interest.

References

[1] A. Blair, B. Ritz, C. Wesseling, L.B. Freeman, Pesticides and human health, Occup. Environ. Med. 72 (2015) 81–82, https://doi.org/10.1136/oemed-2014-102454.
[2] N. Jahan Pinky, S. Mohidul Islam, R. Sharmin Alice, Edibility detection of mushroom using ensemble methods, image, Graph. Signal Process. 4 (2019) 55–62, https://doi.org/10.5815/ijigsp.2019.04.05.
[3] E. Boin, J. Nunes, Mushroom consumption behavior and influencing factors in a sample of the Portuguese population, J. Int. Food Agribus. Mark. 30 (2018) 35–48, https://doi.org/10.1080/08974438.2017.1382420.
[4] J. Azieana, M.N. Zainon, A. Noriham, M.N. Rohana, Morphological identification and toxicity evaluation of two selected wild mushrooms, Adv. Environ. Biol. 9 (2015) 6–10. ⟨https://www.researchgate.net/publication/332553356_Morphological_identification_and_toxicity_evaluation_of_two_selected_wild_mushrooms⟩ (accessed September 16, 2021).
[5] B. Stachowiak, J. Reguła, Health-promoting potential of edible macromycetes under special consideration of polysaccharides: a review, 2012, 369–380. https://doi.org/10.1007/s00217–011-1656–9.
[6] & Georgios, I.I. Zervakis, Medicinal mushrooms: valuable biological resources of high exploitation potential, Plant Biosyst. Int. J. Deal. All Asp. Plant Biol. 151 (2017) 548–565, https://doi.org/10.1080/11263504.2017.1301590.
[7] N. Nowacka, R. Nowak, M. Drozd, M. Olech, R. Los, A. Malm, Antibacterial, Antiradical Potential and Phenolic Compounds of Thirty-One Polish Mushrooms, 2015. https://doi.org/10.1371/journal.pone.0140355.
[8] A. Sabarwal, K. Kumar, R.P. Singh, Hazardous effects of chemical pesticides on human health–Cancer and other associated disorders, Environ. Toxicol. Pharmacol. 63 (2018) 103–114, https://doi.org/10.1016/j.etap.2018.08.018.
[9] H. Rathore, S. Prasad, S. Sharma, Mushroom nutraceuticals for improved nutrition and better human health: a review, PharmaNutrition 5 (2017) 35–46, https://doi.org/10.1016/j.phanu.2017.02.001.
[10] B. Stachowiak, J. Reguła, Health-promoting potential of edible macromycetes under special consideration of polysaccharides: a review, n.d. https://doi.org/10.1007/s00217–011-1656–9.
[11] D.M. Ba, X. Gao, L. Al-Shaar, J. Muscat, V.M. Chinchilli, P. Ssentongo, X. Zhang, G. Liu, R.B. Beelman, J.P. Richie, Prospective study of dietary mushroom intake and risk of mortality: results from continuous National Health and Nutrition Examination Survey (NHANES) 2003-2014 and a meta-analysis, Nutr. J. 20 (2021) 1–12, https://doi.org/10.1186/s12937-021-00738-w.
[12] D. Kothari, S. Patel, S.K. Kim, Anticancer and other therapeutic relevance of mushroom polysaccharides: a holistic appraisal, Biomed. Pharmacother. 105 (2018) 377–394, https://doi.org/10.1016/j.biopha.2018.05.138.
[13] M. Gąsecka, M. Siwulski, M. Mleczek, Evaluation of bioactive compounds content and antioxidant properties of soil-growing and wood-growing edible mushrooms, J. Food Process. Preserv. 42 (2018) 1–10, https://doi.org/10.1111/jfpp.13386.
[17] A. Zhang, X. Li, C. Xing, J. Yang, P. Sun, Antioxidant activity of polysaccharide extracted from *Pleurotus eryngii* using Response surface methodology, Int. J. Biol. Macromol. (2014), https://doi.org/10.1016/j.ijbiomac.2014.01.013.
[18] G. Aa, S. Kr, Pavithra M., Functional attributes of ethnically edible ectomycorrhizal wild mushroom *Amanita* in India, (2018). ⟨http://fungiofegypt.com/Journal/index.html⟩ (accessed August 9, 2021).
[19] N. Gursoy, C. Sarikurkcu, M. Cengiz, M.H. Solak, Antioxidant activities, metal contents, total phenolics and flavonoids of seven *Morchella* species, (n.d.). https://doi.org/10.1016/j.fct.2009.06.032.
[20] T. Ozen, C. Darcan, O. Aktop, I. Turkekul, Screening of antioxidant, antimicrobial activities and chemical contents of edible mushrooms wildly grown in the Black Sea Region of Turkey, Comb. Chem. High. Throughput Screen. 14 (2011) 72–84.
[21] H.Y.Y. Yap, M.J. Muria-Gonzalez, B.H. Kong, K.A. Stubbs, C.S. Tan, S.T. Ng, N. H. Tan, P.S. Solomon, S.Y. Fung, Y.H. Chooi, Heterologous expression of cytotoxic sesquiterpenoids from the medicinal mushroom *Lignosus rhinocerotis* in yeast, Microb. Cell Fact. 16 (2017) 1–13, https://doi.org/10.1186/s12934-017-0713-x.
[22] M. Fogarasi, S.A. Socaci, F.V. Dulf, Z.M. Diaconeasa, A.C. Fărcas, M. Tofană, C. A. Semeniuc, Bioactive compounds and volatile profiles of five transylvanian wild edible mushrooms, Molecules 23 (2018), https://doi.org/10.3390/molecules23123272.
[23] F. Kalyoncu, M. Oskay, H. Kayalar, Antioxidant activity of the mycelium of 21 wild mushroom species, Mycology 1 (2010) 195–199, https://doi.org/10.1080/21501203.2010.511292.
[24] Z.M. Thu, K. Ko Myo, H.T. Aung, M. Clericuzio, C. Armijos, G. Vidari, Bioactive phytochemical constituents of wild edible mushrooms from Southeast Asia, Molecules 25 (2020), https://doi.org/10.3390/molecules25081972.
[25] M.-Y. Yeh, W.-C. Ko, L.-Y. Lin, Hypolipidemic and Antioxidant Activity of Enoki Mushrooms (*Flammulina velutipes*), (2014). https://doi.org/10.1155/2014/352385.
[26] K. Puukka, J. Aho, A. Mustonen, M. Maija, S. Saarela, P. Nieminen, Myo- and cardiotoxic effects of the wild winter mushroom (*Flammulina velutipes*) on mice, Exp. Biol. Med. 243 (2018) 639–644, https://doi.org/10.1177/1535370218762340.
[27] J.P. Kehrer, L.O. Klotz, Free radicals and related reactive species as mediators of tissue injury and disease: Implications for Health, Crit. Rev. Toxicol. 45 (2015) 765–798, https://doi.org/10.3109/10408444.2015.1074159.
[28] S. Askarova, B. Umbayev, A. Masoud, A. Kaiyrlykyzy, The links between the gut microbiome, aging, modern lifestyle and Alzheimer's disease, Front. Cell. Infect. Microbiol. 10 (2020) 1–12, https://doi.org/10.3389/fcimb.2020.00104.
[29] M. Widel, A. Krzywon, K. Gajda, M. Skonieczna, J. Rzeszowska-Wolny, Induction of bystander effects by UVA, UVB, and UVC radiation in human fibroblasts and the implication of reactive oxygen species, Free Radic. Biol. Med. 68 (2014) 278–287, https://doi.org/10.1016/j.freeradbiomed.2013.12.021.
[30] M.P. Barros, M.J. Rodrigo, L. Zacarias, Dietary carotenoid roles in redox homeostasis and human health, J. Agric. Food Chem. 66 (2018) 5733–5740, https://doi.org/10.1021/acs.jafc.8b00866.
[31] F. Rodrigues, M.A. da Mota Nunes, M.B.P. Pinto Oliveira, Applications of recovered bioactive compounds in cosmetics and health care products, in: Olive Mill Waste Recent Adv. Sustain. Manag., Elsevier Inc., 2017, pp. 255–274, https://doi.org/10.1016/B978-0-12-805314-0.00012-1.
[32] T. Ao, C.R. Deb, Nutritional and antioxidant potential of some wild edible mushrooms of Nagal and, India, J. Food Sci. Technol. 56 (2019) 1084–1089, https://doi.org/10.1007/s13197-018-03557-w.
[33] D.A. Abugri, W.H. Mcelhenney, Extraction of total phenolic and flavonoids from edible wild and cultivated medicinal mushrooms as affected by different solvents, J. Nat. Prod. Plant Resour. 3 (2013) 37–42. ⟨http://scholarsresearchlibrary.com/archive.html⟩.
[34] E. Enow Andrew, T. Rosemary Kinge, E. Maureen Tabi, N. Thiobal, A. Mathias Mih, Diversity and distribution of macrofungi (mushrooms) in the Mount Cameroon Region, J. Ecol. Nat. Environ. 5 (2013) 318–334, https://doi.org/10.5897/JENE2013.0379.
[35] R.W. Mwangi, J.M. Macharia, I.N. Wagara, R.L. Bence, The medicinal properties of *Cassia fistula* L: a review, Biomed. Pharmacother. 144 (2021) 1–9, https://doi.org/10.1016/j.biopha.2021.112240.
[36] T. Devasagayam, J. Tilak, K. Boloor, K.S. Sane, S.S. Ghaskadbi, R. Lele, Free radicals and antioxidants in human health: current status and future prospects, J. Appl. Physiol. 52 (2004) 794–804, https://doi.org/10.1142/s0218843097000161.
[37] D. Prakash, C. Gupta, G. Sharma, Importance of phytochemicals in nutraceuticals, J. Chin. Med. Res. Dev. 1 (2012) 70–78. ⟨https://www.researchgate.net/publication/281369030⟩.

[38] S. Wasser, Medicinal mushrooms as a source of antitumor and immunomodulating polysaccharides, Appl. Microbiol. Biotechnol. 60 (2002) 258–274, https://doi.org/10.1007/s00253-002-1076-7.
[39] S.P. Wasser, A. Weis, Therapeutic effects of substances occurring in higher Basidiomycetes mushrooms: a modern perspective, Crit. Rev. Immunol. 19 (1999) 65–96, https://doi.org/10.1615/CritRevImmunol.v19.i1.30.
[40] M. Kozarski, A. Klaus, J. Vunduk, Z. Zizak, M. Niksic, D. Jakovljevic, M.M. Vrvic, L.J.L.D. Van Griensven, Nutraceutical properties of the methanolic extract of edible mushroom *Cantharellus cibarius* (Fries): primary mechanisms, Food Funct. 6 (2015) 1875–1886, https://doi.org/10.1039/c5fo00312a.
[41] V. Loria-Kohen, T. Lourenço-Nogueira, I. Espinosa-Salinas, F.R. Marín, C. Soler-Rivas, A.R. de Molina, Nutritional and functional properties of edible mushrooms: a food with promising health claims, J. Pharm. Nutr. Sci. 4 (2014) 187–198, https://doi.org/10.6000/1927-5951.2014.04.03.4.
[42] D.M. Ba, X. Gao, J. Muscat, L. Al-Shaar, V. Chinchilli, X. Zhang, P. Ssentongo, R. B. Beelman, J.P. Richie, Association of mushroom consumption with all-cause and cause-specific mortality among American adults: prospective cohort study findings from NHANES III, Nutr. J. 20 (2021) 1–11, https://doi.org/10.1186/s12937-021-00691-8.
[43] J.M. Pinto, V. Merzbach, A.G.B. Willmott, J. Antonio, J. Roberts, Assessing the impact of a mushroom-derived food ingredient on vitamin D levels in healthy volunteers, J. Int. Soc. Sports Nutr. 17 (2020) 1–10, https://doi.org/10.1186/s12970-020-00387-0.
[44] M.S. Brewer, Natural antioxidants: sources, compounds, mechanisms of action, and potential applications, Compr. Rev. Food Sci. Food Saf. 10 (2011) 221–247, https://doi.org/10.1111/j.1541-4337.2011.00156.x.
[45] M. Kozarski, A. Klaus, D. Jakovljevic, N. Todorovic, J. Vunduk, P. Petrović, M. Niksic, M.M. Vrvic, L. Van Griensven, Antioxidants of edible mushrooms, Molecules 20 (2015) 19489–19525, https://doi.org/10.3390/molecules201019489.
[46] A. Klaus, M. Kozarski, M. Niksic, D. Jakovljevic, N. Todorovic, L.J.L.D. Van Griensven, Antioxidative activities and chemical characterization of polysaccharides extracted from the basidiomycete *Schizophyllum commune*, LWT - Food Sci. Technol. 44 (2011) 2005–2011, https://doi.org/10.1016/j.lwt.2011.05.010.
[47] M. Kozarski, A. Klaus, M. Niksic, D. Jakovljevic, J.P.F.G. Helsper, L.J.L.D. Van Griensven, Antioxidative and immunomodulating activities of polysaccharide extracts of the medicinal mushrooms *Agaricus bisporus, Agaricus brasiliensis, Ganoderma lucidum* and *Phellinus linteus*, Food Chem. 129 (2011) 1667–1675, https://doi.org/10.1016/j.foodchem.2011.06.029.
[48] F.S. Reis, A. Martins, L. Barros, I.C.F.R. Ferreira, Antioxidant properties and phenolic profile of the most widely appreciated cultivated mushrooms: a comparative study between in vivo and in vitro samples, Food Chem. Toxicol. 50 (2012) 1201–1207, https://doi.org/10.1016/j.fct.2012.02.013.
[49] C. YouGuo, S. ZongJi, C. XiaoPing, Modulatory effect of *Ganoderma lucidum* polysaccharides on serum antioxidant enzymes activities in ovarian cancer rats, Carbohydr. Polym. 78 (2009) 258–262, https://doi.org/10.1016/j.carbpol.2009.03.030.
[50] K.R. Martin, Both common and specialty mushrooms inhibit adhesion molecule expression and in vitro binding of monocytes to human aortic endothelial cells in a pro-inflammatory environment, Nutr. J. 9 (2010) 1–9, https://doi.org/10.1186/1475-2891-9-29.
[51] A. Dogan, Abdullah Dalar, Cemil Sadullahoglu, Abdulhamit Battal, Yusuf Uzun, I. Celik, K. Demirel, Investigation of the protective effects of horse mushroom (*Agaricus arvensis* Schaeff.) against carbon tetrachloride-induced oxidative stress in rats, 1 (n.d.) 3. https://doi.org/10.1007/s11033–018-4218–4.
[52] J. Liu, L. Jia, J. Kan, C.-H. Jin, In vitro and in vivo antioxidant activity of ethanolic extract of white button mushroom (*Agaricus bisporus*), Food Chem. Toxicol. 51 (2013) 310–316, https://doi.org/10.1016/j.fct.2012.10.014.
[53] F. Mourão, S.H. Umeo, O.S. Takemura, A. Linde, N.B. Colauto, Antioxidant activity of *Agaricus brasiliensis* basidiocarps on different maturation phases, Braz. J. Microbiol. 42 (2011) 197–202.
[54] E. Pereira, L. Barros, A. Martins, I.C.F.R. Ferreira, Towards chemical and nutritional inventory of Portuguese wild edible mushrooms in different habitats, Food Chem. 130 (2012) 394–403, https://doi.org/10.1016/j.foodchem.2011.07.057.
[55] M. Kosanić, B. Ranković, A. Rančić, T. Stanojković, Evaluation of metal contents and bioactivity of two edible mushrooms *Agaricus campestris* and *Boletus edulis*, Emir. J. Food Agric. 29 (2017), https://doi.org/10.9755/ejfa.2016-06-656.
[56] L. Barros, S. Falcão, P. Baptista, C. Freire, M. Vilas-Boas, I.C.F.R. Ferreira, Antioxidant activity of *Agaricus* sp. mushrooms by chemical, biochemical and electrochemical assays, (n.d.). https://doi.org/10.1016/j.foodchem.2008.03.033.
[57] L.W. Gao, J.W. Wang, Antioxidant potential and dna damage protecting activity of aqueous extract from *Armillaria mellea*, J. Food Biochem. 36 (2012) 139–148, https://doi.org/10.1111/J.1745-4514.2010.00520.X.
[58] Q. Junqiang, H. Zhang, W. Zhenyu, Ultrasonic degradation of Polysaccharides from *Auricularia auricula* and the antioxidant activity of their degradation products, LWT - Food Sci. Technol. (2019) 1–8, 10.1016/j.lwt.2019.108266 (accessed August 9, 2021).
[59] Y.-C. Hseu, S.-C. Chen, Y.-J. Yech, L. Wang, H.-L. Yang, Antioxidant activity of *Antrodia camphorata* on free radical-induced endothelial cell damage, J. Ethnopharmacol. 118 (2008) 237–245, https://doi.org/10.1016/j.jep.2008.04.004.
[60] G. Jaworska, K. Pogoń, A. Skrzypczak, E. Bernaś, Composition and antioxidant properties of wild mushrooms *Boletus edulis* and *Xerocomus badius* prepared for consumption, J. Food Sci. Technol. 52 (2015) 7944–7953, https://doi.org/10.1007/s13197-015-1933-x.
[61] M. Sevindik, Wild edible mushroom *Cantharellus cibarius* as a natural antioxidant food Doğal bir Antioksidan Gıda Olarak Yenilebilir Yabani Mantar Cantharellus cibarius, Turk. J. Agric. - Food Sci. Technol. 7 (2019) 1377–1381.
[62] O.O. Ogbole, A.O. Nkumah, A.U. Linus, M.O. Falade, Molecular identification, in vivo and in vitro activities of *Calvatia gigantea* (macro-fungus) as an antidiabetic agent, Mycology 10 (2019) 166–173, https://doi.org/10.1080/21501203.2019.1595204.
[63] P. Hong Leung, S. Zhao, K. Ping Ho, J. Yong Wu, Chemical properties and antioxidant activity of exopolysaccharides from mycelial culture of *Cordyceps sinensis* fungus Cs-HK1, Food Chem. 114 (2009) 1251–1256, https://doi.org/10.1016/j.foodchem.2008.10.081.
[64] J. Han, Y. Chen, L. Bao, X. Yang, D. Liu, S. Li, F. Zhao, H. Liu, Anti-inflammatory and cytotoxic cyathane diterpenoids from the medicinal fungus *Cyathus africanus*, Fitoterapia 84 (2013) 22–31, https://doi.org/10.1016/j.fitote.2012.10.001.
[65] Y. Liu, B. Zhang, S.A. Ibrahim, S.S. Gao, H. Yang, W. Huang, Purification, characterization and antioxidant activity of polysaccharides from *Flammulina velutipes* residue, Carbohydr. Polym. 145 (2016) 71–77, https://doi.org/10.1016/j.carbpol.2016.03.020.
[66] M. Kozarski, A. Klaus, N. Nikšićá, M.M. Vrvicb, V. Vrvicb, N. Todorovicć, T. Todorovicć, D. Jakovljevicć, J. Jakovljevicć, L.J.L.D. Van Griensven, Antioxidative activities and chemical characterization of polysaccharide extracts from the widely used mushrooms *Ganoderma applanatum, Ganoderma lucidum, Lentinus edodes* and *Trametes versicolor*, J. Food Compos. Anal. 26 (2012) 144–153, https://doi.org/10.1016/j.jfca.2012.02.004.
[67] M. Shi, Z. Zhang, Y. Yang, Antioxidant and immunoregulatory activity of *Ganoderma lucidum* polysaccharide (GLP), Carbohydr. Polym. 95 (2013) 200–206, https://doi.org/10.1016/j.carbpol.2013.02.081.
[68] Z.H. Han, J.M. Ye, G.F. Wang, Evaluation of in vivo antioxidant activity of *Hericium erinaceus* polysaccharides, Int. J. Biol. Macromol. 52 (2013) 66–71, https://doi.org/10.1016/j.ijbiomac.2012.09.009.
[69] A.A.J. Carneiro, I.C.F.R. Ferreira, M. Dueñas, L. Barros, R. Da Silva, E. Gomes, C. Santos-Buelga, Chemical composition and antioxidant activity of dried powder formulations of *Agaricus blazei* and *Lentinus edodes*, Food Chem. 138 (2013) 2168–2173, https://doi.org/10.1016/j.foodchem.2012.12.036.
[70] B.F. Lau, N. Abdullah, N. Aminudin, H.B. Lee, P.J. Tan, Ethnomedicinal uses, pharmacological activities, and cultivation of *Lignosus* spp. (tigeŕs milk mushrooms) in Malaysia - a review, J. Ethnopharmacol. 169 (2015) 441–458, https://doi.org/10.1016/j.jep.2015.04.042.
[71] L. Antigua Guatemala, Chemical composition and bioactivity of wild samples of *Morchella esculenta* from Portugal and Serbia., in: 2017: p. 2017.
[72] P. He, F. Li, L. Huang, D. Xue, W. Liu, C. Xu, Chemical characterization and antioxidant activity of polysaccharide extract from spent mushroom substrate of *Pleurotus eryngii*, J. Taiwan Inst. Chem. Eng. 69 (2016) 48–53, https://doi.org/10.1016/j.jtice.2016.10.017.
[73] S. Khatun, A. Islam, U. Cakilcioglu, P. Guler, N.C. Chatterjee, Nutritional qualities and antioxidant activity of three edible oyster mushrooms (*Pleurotus* spp.), NJAS - Wagening. J. Life Sci. 72 (2015) 1–5, https://doi.org/10.1016/j.njas.2012.03.003.
[74] P.A. Fontes Vieira, D.C. Gontijo, B.C. Vieira, E.A.F. Fontes, L.S. de Assunção, J.P. V. Leite, M.G. de, A. Oliveira, M.C.M. Kasuya, Antioxidant activities, total phenolics and metal contents in *Pleurotus ostreatus* mushrooms enriched with iron, zinc or lithium, LWT - Food Sci. Technol. 54 (2013) 421–425, https://doi.org/10.1016/J.LWT.2013.06.016.
[75] T.C. Finimundy, G. Gambato, R. Fontana, M. Camassola, M. Salvador, S. Moura, J. Hess, J.A.P. Henriques, A.J.P. Dillon, M. Roesch-Ely, Aqueous extracts of *Lentinula edodes* and *Pleurotus sajor-caju* exhibit high antioxidant capability and promising in vitro antitumor activity, Nutr. Res. 33 (2013) 76–84, https://doi.org/10.1016/j.nutres.2012.11.005.
[76] Â. Fernandes, J. Petrović, D. Stojković, L. Barros, J. Glamočlija, M. Soković, A. Martins, I.C.F.R. Ferreira, *Polyporus squamosus* (Huds.) Fr from different origins: chemical characterization, screening of the bioactive properties and specific antimicrobial effects against *Pseudomonas aeruginosa*, LWT - Food Sci. Technol. 69 (2016) 91–97, https://doi.org/10.1016/j.lwt.2016.01.037.
[77] A. Omarini, C. Henning, J. Ringuelet, J.A. Zygadlo, E. Albertó, Volatile composition and nutritional quality of the edible mushroom *Polyporus tenuiculus* grown on different agro-industrial waste, Int. J. Food Sci. Technol. 45 (2010) 1603–1609, https://doi.org/10.1111/j.1365-2621.2010.02306.x.
[78] N. Gursoy, C. Sarikurkcu, B. Tepe, M.H. Solak, Evaluation of antioxidant activities of 3 edible mushrooms: *Ramaria flava* (Schaef.: Fr.) Quél., Rhizopogon roseolus (Corda) T.M. Fries., and Russula delica Fr, Food Sci. Biotechnol. 19 (2010) 691–696, https://doi.org/10.1007/s10068-010-0097-8.
[79] H. Li, X. Wang, Q. Xiong, Y. Yu, L. Peng, Sulfated modification, characterization, and potential bioactivities of polysaccharide from the fruiting bodies of *Russula virescens*, Int. J. Biol. Macromol. 154 (2020) 1438–1447, https://doi.org/10.1016/j.ijbiomac.2019.11.025.
[80] M. Elmastas, O. Isildak, I. Turkekul, N. Temur, Determination of antioxidant activity and antioxidant compounds in wild edible mushrooms, J. Food Compos. Anal. 20 (2007) 337–345, https://doi.org/10.1016/j.jfca.2006.07.003.
[81] L.M. Cheung, P.C.K. Cheung, Mushroom extracts with antioxidant activity against lipid peroxidation, Food Chem. 89 (2005) 403–409, https://doi.org/10.1016/j.foodchem.2004.02.049.
[82] K. Canli, I. Akata, E.M. Altuner, *In vitro* antimicrobial activity screening of *Xylaria hypoxylon*, Afr. J. Tradit. Complement. Altern. Med 13 (2016) 42–46, https://doi.org/10.21010/ajtcam.v13i4.7.

[83] V.I. Lushchak, Free radicals, reactive oxygen species, oxidative stress and its classification, Chem. Biol. Interact. 224 (2014) 164–175, https://doi.org/10.1016/j.cbi.2014.10.016.
[84] V. Lobo, A. Patil, A. Phatak, N. Chandra, Free radicals, antioxidants and functional foods: impact on human health, Pharmacogn. Rev. 4 (2010) 118–126, https://doi.org/10.4103/0973-7847.70902.
[85] C. Ma-on, A. Sanpavat, P. Whongsiri, S. Suwannasin, N. Hirankarn, P. Tangkijvanich, C. Boonla, Oxidative stress indicated by elevated expression of Nrf2 and 8-OHdG promotes hepatocellular carcinoma progression, Med. Oncol. 34 (2017) 904–919, https://doi.org/10.1007/s12032-017-0914-5.
[86] I. Liguori, G. Russo, F. Curcio, G. Bulli, L. Aran, D. Della-Morte, G. Gargiulo, G. Testa, F. Cacciatore, D. Bonaduce, P. Abete, Oxidative stress, aging, and diseases, Clin. Interv. Aging 13 (2018) 757–772, https://doi.org/10.2147/CIA.S158513.
[87] K.A. Zaman, A.A. Khalid, Free radical scavenging activity of some bangladeshi medicinal plants, Pharmacologyonline 3 (2015) 29–32.
[88] M.A. Hasnat, M. Pervin, T. Debnath, B.O. Lim, DNA protection, total phenolics and antioxidant potential of the mushroom *Russula virescens*, J. Food Biochem. 38 (2014) 6–17, https://doi.org/10.1111/jfbc.12019.
[89] S.-Y. Fung, C.-S. Tan, The Bioactivity of Tiger Milk Mushroom: Malaysia's Prized Medicinal Mushroom, in: 2017: pp. 111–133. https://doi.org/10.1007/978–981-10–5978-0_5.
[90] O. Yildiz, Z. Can, A.Q. Laghari, H. Şahin, M. Malkoç, Wild edible mushrooms as a natural source of phenolics and antioxidants, J. Food Biochem. 39 (2015) 148–154, https://doi.org/10.1111/jfbc.12107.
[91] J.L. Mau, H.C. Lin, C.C. Chen, Antioxidant properties of several medicinal mushrooms, J. Agric. Food Chem. 50 (2002) 6072–6077, https://doi.org/10.1021/jf0201273.
[92] M. Mleczek, P. Rzymski, A. Budka, M. Siwulski, A. Jasińska, P. Kalač, B. Poniedziałek, M. Gąsecka, P. Niedzielski, Elemental characteristics of mushroom species cultivated in China and Poland, J. Food Compos. Anal. 66 (2018) 168–178, https://doi.org/10.1016/j.jfca.2017.12.018.
[93] A. Mocan, Â. Fernandes, L. Barros, G. Crişan, M. Smiljković, M. Soković, I.C.F.R. Ferreira, Chemical composition and bioactive properties of the wild mushroom *Polyporus squamosus* (Huds.) Fr: a study with samples from Romania, 9 (2018) 160. https://doi.org/10.1039/c7fo01514c.
[94] D. Menaga, S. Rajakumar, P.M. Ayyasamy, Free radical scavenging activity of methanolic extract of *Pleurotus florida* mushroom, Int. J. Pharm. Pharm. Sci. 5 (2013) 601–606. ⟨https://www.researchgate.net/publication/266798248⟩. accessed August 16, 2021.
[95] A. R, B.P. R, Protection of natural antioxidants against low-density lipoprotein oxidation. Advances in Food and Nutrition Research | 10.1016/bs.afnr.2020.04.002, Adv. Food Nutr. Res. 23 (2017) 2754–2766.10.1016/bs.afnr.2020.04.002 (accessed August 16, 2021).
[96] I. Milovanović, M. Stajić, J. Ćilerdžić, T. Stanojković, A. Knežević, J. Vukojević, Antioxidant, antifungal and anticancer activities of Se-enriched *Pleurotus* spp. mycelium extracts, Arch. Biol. Sci. 66 (2014) 1379–1388, https://doi.org/10.2298/ABS1404379M.
[97] J.M. Hussein, D.D. Tibuhwa, A.M. Mshandete, A.K. Kivaisi, Antioxidant properties of seven wild edible mushrooms from Tanzania, Afr. J. Food Sci. 9 (2015) 471–479, https://doi.org/10.5897/ajfs2015.1328.
[98] N. Abdullah, S.M. Ismail, N. Aminudin, A.S. Shuib, B.F. Lau, Evaluation of selected culinary-medicinal mushrooms for antioxidant and ACE inhibitory activities, Evid.-Based Complement. Altern. Med. 2012 (2012) 1–12, https://doi.org/10.1155/2012/464238.
[100] H.P. V. Nhi, HoChiMinh City, Quarter 6, Linh Trung Ward, 2012.
[101] D. K, S. M, Pleurotus ostreatus: an oyster mushroom with nutritional and medicinal properties, J. Biochem. Technol. 5, 2014, 718–726. moz-extension://198bafd1–6951-4e62–8bca-cdec4c5a0caa/enhanced-reader.html?openApp&pdf=https%3A%2F%2Fwww.klmtechgroup.com%2FPDF%2FEDG-REF%2FENGINEERING-DESIGN-GUIDELINES-vacuum-tower-selection-and-sizing-Rev2.3web.pdf (accessed August 16, 2021).
[102] C. Sánchez, Reactive oxygen species and antioxidant properties from mushrooms, Synth. Syst. Biotechnol. 2 (2017) 13–22, https://doi.org/10.1016/j.synbio.2016.12.001.
[103] İ. lkay Koca, Ali Keleş, H. Gençcelep, Antioxidant properties of wild edible mushrooms, J. Food Process. Technol. 02 (2011), https://doi.org/10.4172/2157-7110.1000130.
[104] K. Sukalingam, K. Ganesan, B. Xu, *Trianthema portulacastrum* L. (giant pigweed): phytochemistry and pharmacological properties, Phytochem. Rev. 16 (2017) 461–478, https://doi.org/10.1007/s11101-017-9493-5.
[105] C.V. Toledo, C. Barroetaveñ, Â. Fernandes, L. Barros, I.C.F.R. Ferreira, Chemical and antioxidant properties of wild edible mushrooms from native nothofagus spp. forest, Argentina, Molecules 21 (2016) 1–15, https://doi.org/10.3390/molecules21091201.
[106] K. Manimaran, G. Balasubramani, C. Ragavendran, D. Natarajan, S. Murugesan, Biological applications of synthesized ZnO nanoparticles using *Pleurotus djamor* against mosquito larvicidal, histopathology, antibacterial, antioxidant and anticancer effect, J. Clust. Sci. 32 (2021) 1635–1647, https://doi.org/10.1007/s10876-020-01927-z.
[107] P. Shah, H.A. Modi, Comparative study of DPPH, ABTS and FRAP assays for determination of antioxidant activity, Int. J. Res. Appl. Sci. Eng. Technol. 3 (2015) 2321–9653.
[108] X. Li, Z. Wang, L. Wang, E. Walid, H. Zhang, In vitro antioxidant and anti-proliferation activities of polysaccharides from various extracts of different mushrooms, Int. J. Mol. Sci. 13 (2012) 5801–5817, https://doi.org/10.3390/ijms13055801.
[109] M.K. Panda, S.K. Das, S. Mohapatra, P.R. Debata, K. Tayung, H. Thatoi, Mycochemical composition, bioactivities, and phylogenetic placement of three wild edible *Russula* species from Northern Odisha, India, Plant Biosyst. 2020 (2020) 1–5, https://doi.org/10.1080/11263504.2020.1813829.
[110] M.L. Gargano, L.J.L.D. van Griensven, O.S. Isikhuemhen, U. Lindequist, G. Venturella, S.P. Wasser, G.I. Zervakis, Medicinal mushrooms: Valuable biological resources of high exploitation potential, Plant Biosyst. 151 (2017) 548–565, https://doi.org/10.1080/11263504.2017.1301590.
[111] T. Herraiz, J. Galisteo, Hydroxyl radical reactions and the radical scavenging activity of β-carboline alkaloids, Food Chem. 172 (2015) 640–649, https://doi.org/10.1016/j.foodchem.2014.09.091.
[112] X. Yang, F. Yan, S. Huang, C. Fu, Antioxidant activities of fractions from longan pericarps, Food Sci. Technol. 34 (2014) 341–345, https://doi.org/10.1590/S0101-20612014005000034.
[113] Y. Zhang, X. Lu, Z. Fu, Z. Wang, J. Zhang, Sulphated modification of a polysaccharide obtained from fresh persimmon (*Diospyros kaki* L.) fruit and antioxidant activities of the sulphated derivatives, Food Chem. 127 (2011) 1084–1090, https://doi.org/10.1016/j.foodchem.2011.01.100.
[114] I. Juránek, D. Nikitovic, D. Kouretas, A.W. Hayes, A.M. Tsatsakis, Biological importance of reactive oxygen species in relation to difficulties of treating pathologies involving oxidative stress by exogenous antioxidants, Food Chem. Toxicol. 61 (2013) 240–247, https://doi.org/10.1016/j.fct.2013.08.074.
[115] Y. Zou, A. Jiang, M. Tian, Extraction optimization of antioxidant polysaccharides from *Auricularia auricula* fruiting bodies, Food Sci. Technol. 35 (2015) 428–433, https://doi.org/10.1590/1678-457X.6712.
[116] M. Karaman, K. Atlagić, A. Novaković, F. Šibul, M. Živić, K. Stevanović, B. Pejin, Fatty acids predominantly affect anti-hydroxyl radical activity and FRAP value: the case study of two edible mushrooms, Antioxidants 8 (2019) 1–12, https://doi.org/10.3390/antiox8100480.
[117] L. Chun-hui, W. Chang-hai, X. Zhi-liang, W. Yi, Isolation, chemical characterization and antioxidant activities of two polysaccharides from the gel and the skin of *Aloe barbadensis* Miller irrigated with sea water, Process Biochem. 42 (2007) 961–970, https://doi.org/10.1016/j.procbio.2007.03.004.
[118] T.Y. Vitak, S.P. Wasser, E. Nevo, N.O. Sybirna, Effect of medicinal mushrooms on L-arginine/NO system in red blood cells of streptozotocin-induced diabetic rats, Adv. Diabetes Metab. 4 (2016) 25–31, https://doi.org/10.13189/adm.2016.040201.
[119] A. Alispahić, A. Šapčanin, M. Salihović, E. Ramić, A. Dedić, M. Pazalja, Phenolic content and antioxidant activity of mushroom extracts from Bosnian market, Bull. Chem. Technol. Bosnia Herzeg. 44 (2015) 5–8.
[120] A.E.S.S. Carvajal, E.A. Koehnlein, A.A. Soares, G.J. Eler, A.T.A. Nakashima, A. Bracht, R.M. Peralta, Bioactives of fruiting bodies and submerged culture mycelia of *Agaricus brasiliensis (A.blazei)* and their antioxidant properties, LWT - Food Sci. Technol. 46 (2012) 493–499, https://doi.org/10.1016/j.lwt.2011.11.018.
[121] J. Pal, S. Ganguly, K.S. Tahsin, K. Acharya, In vitro free radical scavenging activity of wild edible mushroom, *Pleurotus squarrosulus* (Mont.) Singer, Indian J. Exp. Biol. 48 (2010) 1210–1218.
[122] L. Martínez, S. Jongberg, G. Ros, L.H. Skibsted, G. Nieto, Plant derived ingredients rich in nitrates or phenolics for protection of pork against protein oxidation, Food Res. Int. 129 (2020), 108789, https://doi.org/10.1016/j.foodres.2019.108789.
[123] B. Sen Wang, J.H. Chen, Y.C. Liang, P. Der Duh, Effects of Welsh onion on oxidation of low-density lipoprotein and nitric oxide production in macrophage cell line RAW 264.7, Food Chem. 91 (2005) 147–155, https://doi.org/10.1016/j.foodchem.2004.06.009.
[124] N. Abdullah, R. Abdulghani, S.M. Ismail, M.H.Z. Abidin, Immune-stimulatory potential of hot water extracts of selected edible mushrooms, Food Agric. Immunol. 28 (2017) 374–387, https://doi.org/10.1080/09540105.2017.1293011.
[125] J.A. Wagay, D. Vyas, Phenolic quantification and anti-oxidant activity of *Morchella esculenta*, Int. J. Pharma Bio Sci. 2 (2011) 188–197.
[126] A. Bains, A. Tripathi, Evaluation of antioxidant and anti-inflammatory properties of aqueous extract of wild mushrooms collected from Himachal Pradesh, Asian J. Pharm. Clin. Res 10 (2017) 467–472, https://doi.org/10.22159/ajpcr.2017.v10i3.16547.
[127] B. Nitha, S. De, S.K. Adhikari, T.P.A. Devasagayam, K.K. Janardhanan, Evaluation of free radical scavenging activity of morel mushroom, *Morchella esculenta* mycelia: a potential source of therapeutically useful antioxidants, Pharm. Biol. 48 (2010) 453–460, https://doi.org/10.3109/13880200903170789.
[128] A. Baskaran, K.H. Chua, V. Sabaratnam, M. Ravishankar Ram, U.R. Kuppusamy, *Pleurotus giganteus* (Berk. Karun & Hyde), the giant oyster mushroom inhibits NO production in LPS/H2O2 stimulated RAW 264.7 cells via STAT 3 and COX-2 pathways, BMC Complement. Altern. Med. 17 (2017) 1–10, https://doi.org/10.1186/s12906-016-1546-6.
[129] K. Acharya, S. Ghosh, I. Kundu, Pharmacognostic standardization of a well known edible mushroom, *Volvariella volvacea*, J. Appl. Pharm. Sci. 6 (2016) 185–190, https://doi.org/10.7324/JAPS.2016.601129.
[130] N. Nowacka, R. Nowak, M. Drozd, M. Olech, R. Los, A. Malm, Analysis of phenolic constituents, antiradical and antimicrobial activity of edible mushrooms growing wild in Poland, LWT - Food Sci. Technol. 59 (2014) 689–694, https://doi.org/10.1016/j.lwt.2014.05.041.
[131] X. Liu, Y. Chen, L. Wu, X.Q. Wu, Y. Huang, B. Liu, Optimization of polysaccharides extraction from *Dictyophora indusiata* and determination of its

antioxidant activity, Int. J. Biol. Macromol. 103 (2017) 175–181, https://doi.org/10.1016/j.ijbiomac.2017.04.125.

[132] M. Karaman, E. Jovin, R. Malbaša, M. Matavuly, M. Popović, Medicinal and edible *Lignicolous* fungi as natural sources of antioxidative and antibacterial agents, Phyther. Res. 24 (2010) 1473–1481, https://doi.org/10.1002/ptr.2969.

[133] I. Palacios, M. Lozano, C. Moro, M. D'Arrigo, M.A. Rostagno, J.A. Martínez, A. García-Lafuente, E. Guillamón, A. Villares, Antioxidant properties of phenolic compounds occurring in edible mushrooms, Food Chem. 128 (2011) 674–678, https://doi.org/10.1016/j.foodchem.2011.03.085.

[134] M. Valko, D. Leibfritz, J. Moncol, M.T.D. Cronin, M. Mazur, J. Telser, Free radicals and antioxidants in normal physiological functions and human disease, Int. J. Biochem. Cell Biol. 39 (2007) 44–84, https://doi.org/10.1016/j.biocel.2006.07.001.

Further reading

[1] V. Sabaratnam, C.-W. Phan, Discovery and development of neuroprotective agents from natural products neuroactive components of culinary and medicinal mushrooms with potential to mitigate age-related neurodegenerative diseases, Discov. Dev. Neuroprotective Agents Nat. Prod. (2018) 401–413, https://doi.org/10.1016/B978-0-12-809593-5.00010-0.

[2] M.B. Zimmermann, R.F. Hurrell, Nutritional iron defi ciency, www.Thelancet.Com. 370 (2007) 511. ⟨www.thelancet.com⟩ (accessed August 9, 2021).

[3] J.Z. Dong, M.R. Liu, C. Lei, X.J. Zheng, Y. Wang, Effects of Selenium and Light Wavelengths on Liquid Culture of *Cordyceps militaris* Link, (n.d.). https://doi.org/10.1007/s12010–012-9628–5.

[4] C.H. Gan, N.B. Nurul Amira, R. Asmah, Antioxidant analysis of different types of edible mushrooms (*Agaricus bisporous* and *Agaricus brasiliensis*), Int. Food Res. J. 20 (2013) 1095–1102.